EXPERIMENT 9

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Experiment 9 – Design of a PL/SQL Package with Procedures and Shared Cursor (Enterprise Applications)

Experiment
Designing and implementing a PL/SQL package that contains procedures and a shared cursor to fetch and display employee details. This experiment demonstrates modular programming, code reusability, and efficient data handling using packages in PL/SQL.

Aim
To create and implement a PL/SQL package consisting of a package specification and package body that includes procedures and a shared cursor for retrieving and displaying employee data.

Objective

- To understand the concept and structure of PL/SQL packages
- To differentiate between package specification and package body
- To implement procedures inside packages
- To use shared cursors for efficient data retrieval
- To develop modular and reusable database programs

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE) / PostgreSQL

Database Administration Tool:
Oracle SQL Developer / pgAdmin

Problem Statement
In enterprise-level database applications, related operations must be grouped together for better maintainability, performance, and reusability. A PL/SQL package provides a structured

approach to organize procedures and shared cursors, ensuring efficient data retrieval and modular program design.

Practical / Experiment Steps

1. Create an employee table with relevant attributes
2. Insert sample data into the table
3. Create package specification declaring procedures and cursor
4. Create package body defining logic for procedures and cursor
5. Use a shared cursor to fetch employee records
6. Execute the package procedure
7. Display the output

Procedure

1. Open Oracle SQL Developer and connect to the database
2. Create the employee table
3. Insert sample records
4. Create package specification
5. Create package body
6. Compile the package successfully
7. Execute package procedure
8. Display employee details output

Input / Output Details

Input
Table: employee (id, name, gender, salary)

Step-wise Output

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Step 1 – Create Employee Table

```
-- STEP 1: CREATE TABLE
-- Creating employee table with basic details
CREATE TABLE employee (
    emp_id NUMBER PRIMARY KEY,
    emp_name VARCHAR2(50),
    emp_gender VARCHAR2(10),
    emp_salary NUMBER
);
```

Step 2 – Insert Sample Data

```
-- STEP 2: INSERT SAMPLE DATA
-- Inserting updated employee records
INSERT INTO employee VALUES (1, 'Arjun', 'Male', 52000);
INSERT INTO employee VALUES (2, 'Sneha', 'Female', 61000);
INSERT INTO employee VALUES (3, 'Vikram', 'Male', 58000);
INSERT INTO employee VALUES (4, 'Kavya', 'Female', 67000);

COMMIT;
```

Step 3 – Create Package Specification

```
-- STEP 3: PACKAGE SPECIFICATION
CREATE OR REPLACE PACKAGE emp_package AS
    PROCEDURE show_employees;
END emp_package;
/
```

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Step 4 – Create Package Body

```
-- STEP 4: PACKAGE BODY
CREATE OR REPLACE PACKAGE BODY emp_package AS

    CURSOR cur_emp IS
        SELECT emp_id, emp_name, emp_gender, emp_salary FROM employee;

    PROCEDURE show_employees IS
        v_emp_id employee.emp_id%TYPE;
        v_emp_name employee.emp_name%TYPE;
        v_emp_gender employee.emp_gender%TYPE;
        v_emp_salary employee.emp_salary%TYPE;

    BEGIN
        OPEN cur_emp;

        LOOP
            FETCH cur_emp INTO v_emp_id, v_emp_name, v_emp_gender, v_emp_salary;
            EXIT WHEN cur_emp%NOTFOUND;

            DBMS_OUTPUT.PUT_LINE(
                'ID: ' || v_emp_id ||
                ' | Name: ' || v_emp_name ||
                ' | Gender: ' || v_emp_gender ||
                ' | Salary: ' || v_emp_salary
            );
        END LOOP;

        CLOSE cur_emp;

    END show_employees;

END emp_package;
/
```

Step 5 – Execute Package Procedure

```
-- STEP 5: EXECUTE PROCEDURE
BEGIN
    emp_package.show_employees;
END;
/
```

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OUTPUT

Output: 395 ms

```
ID: 1 | Name: Arjun | Gender: Male | Salary: 52000
ID: 2 | Name: Sneha | Gender: Female | Salary: 61000
ID: 3 | Name: Vikram | Gender: Male | Salary: 58000
ID: 4 | Name: Kavya | Gender: Female | Salary: 67000
```

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the structure of PL/SQL packages
- Create package specification and package body
- Use shared cursors inside packages
- Implement modular and reusable database logic
- Apply package concepts in enterprise applications